LS-CAT Management Board Meeting

Keith Brister

Agenda

2 PM Central Time 3 PM Eastern Time

Call In #: 1-888-850-4523
Passcode: 190580

http://conf1.ls-cat.org
http://conf2.ls-cat.org

1. Signing the extension
2. New X-Ray detector
3. New data handling
4. Beamline control software
5. APS-U Status

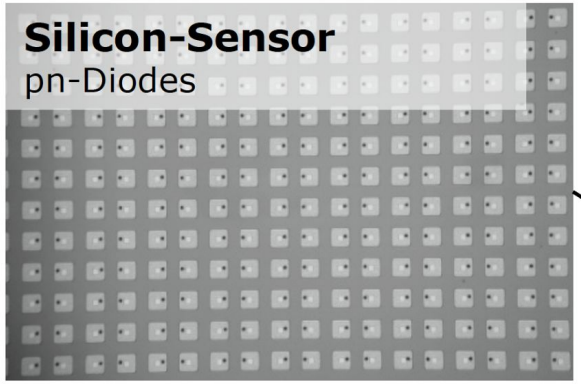
Eiger 9M Arrives at LS-CAT

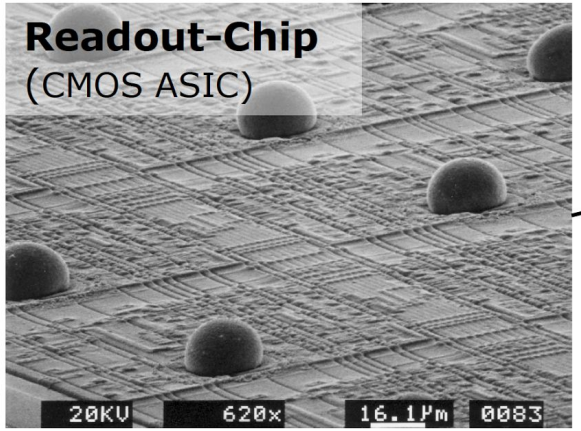
- First large Eiger at APS
- Installation at 21-ID-D starts now
- Hope to see some beam on this detector this run

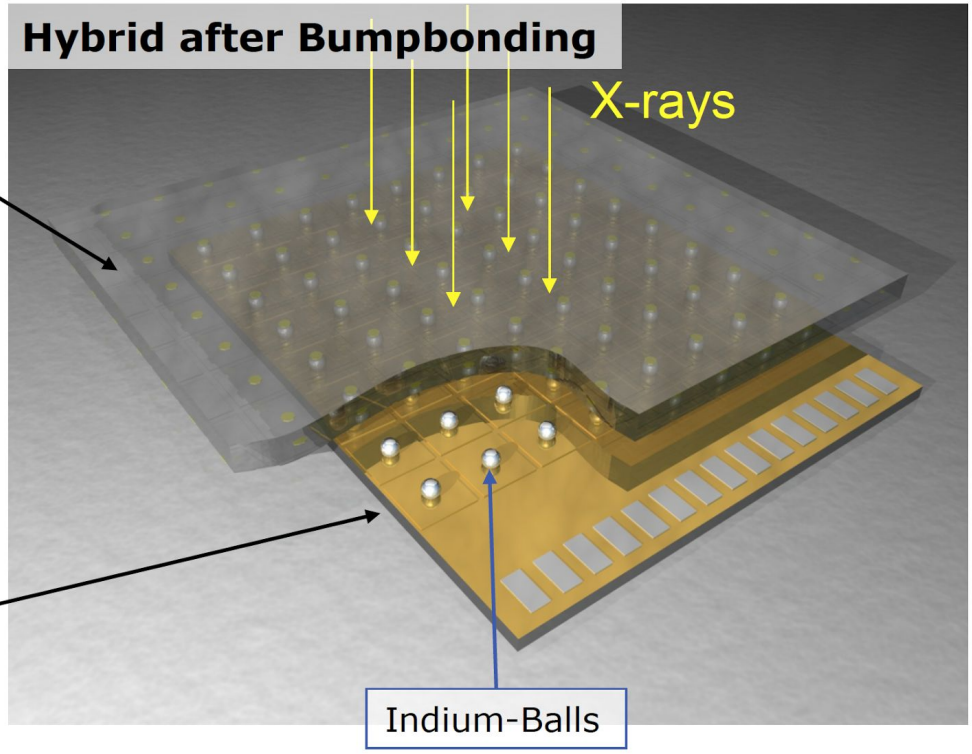




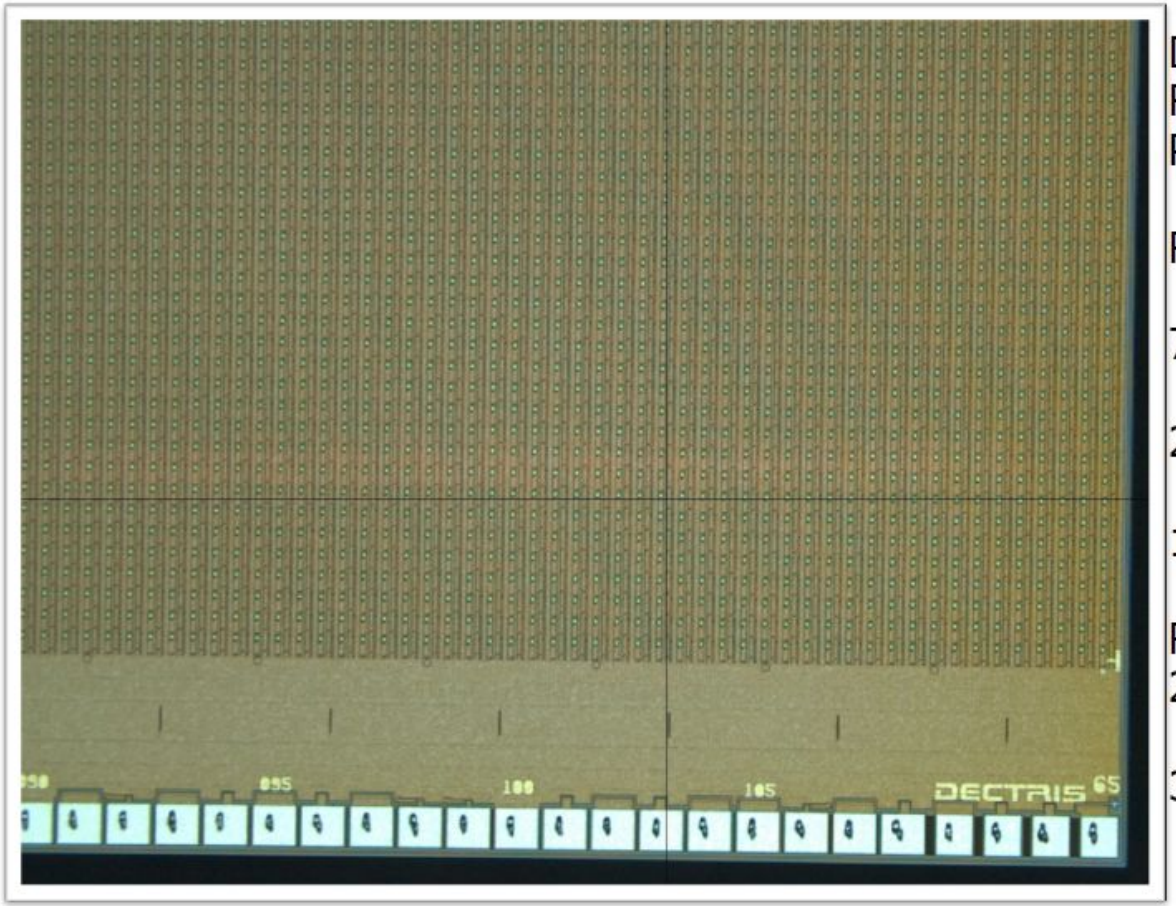
Hybrid-Pixel-Detector: Technology







EIGER – The ASIC



Designed by Roberto Dinapoli, PSI

Radiation tolerant

75 x 75 μm^2 pixel

256 x 256 pixels

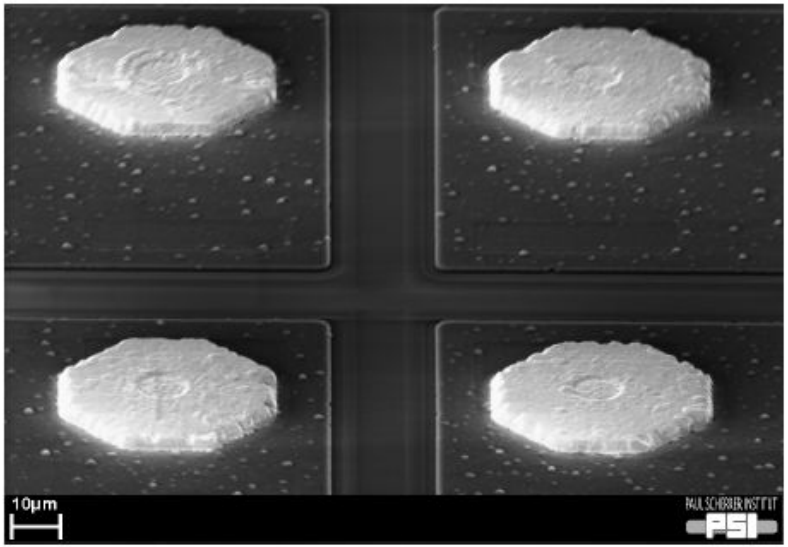
19.3 x 20.1 mm^2

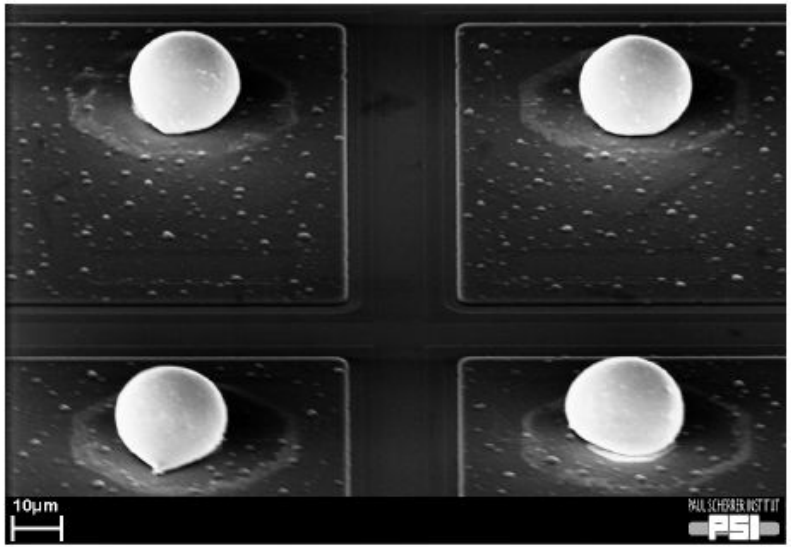
Readout time 240μs

3Gbit/s

Indium Bump-bonding technology

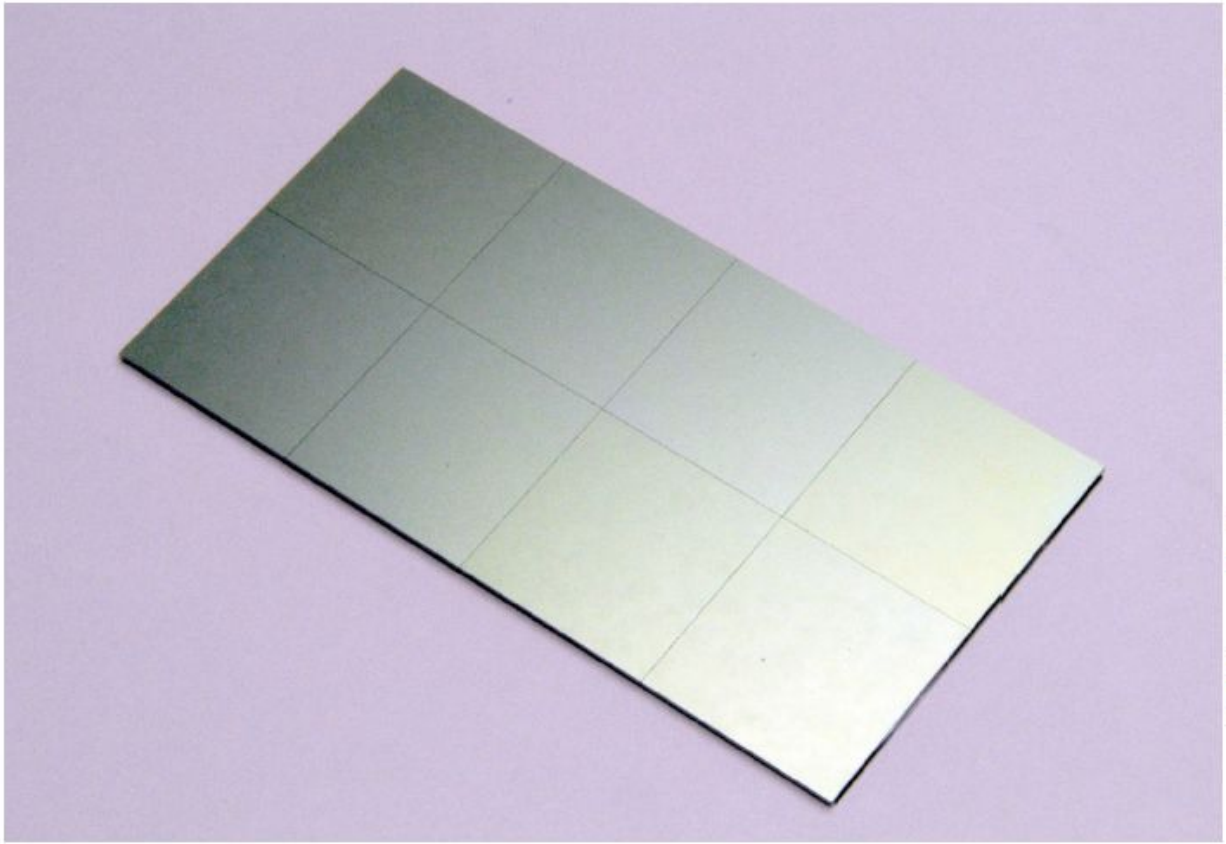
Indium "Cakes" → Indium Bumps





before reflow → after reflow

Eiger Module



8 Chips bump bonded to Silicon Sensor

1030 x 514 = 529420 pixels

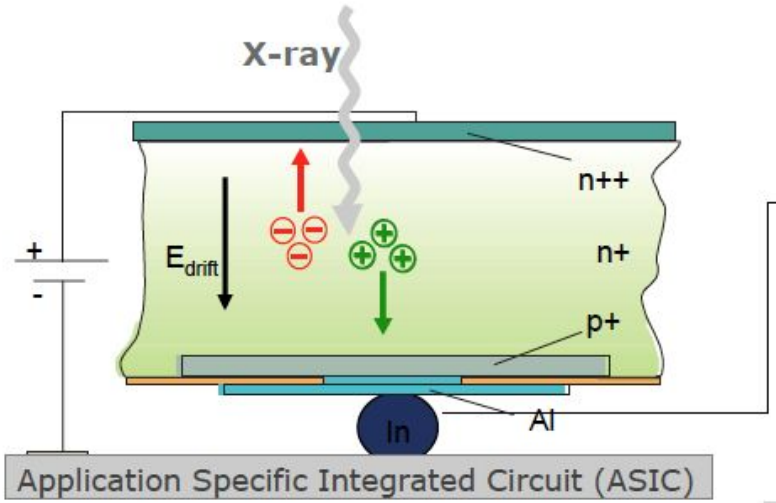
Chips operated in parallel-> 24 Gbits/s

Routinely fabricated by DECTRIS

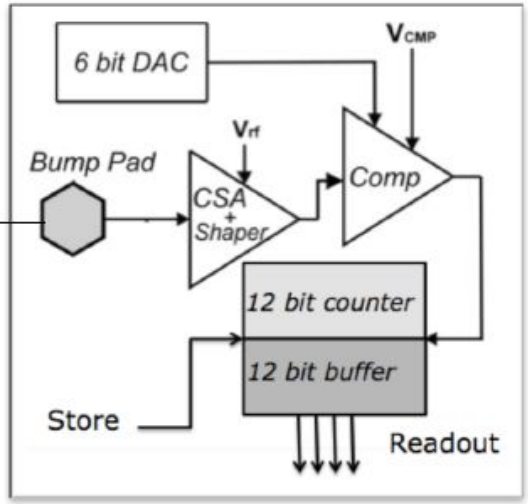
Basic building block of all EIGER Detectors

HPC Detector - Technology

Sensor pixel



Readout pixel



Direct Detection of X-rays in solid state sensor

→ Point Spread Function: 1 pixel

Each pixel is a complete detector

Single Photon Counting in CMOS

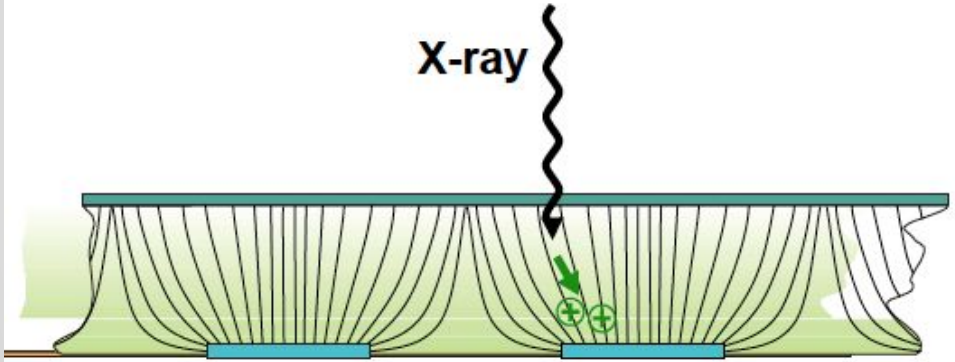
→ no readout noise & dark current

→ adjustable energy threshold

→ high dynamic range

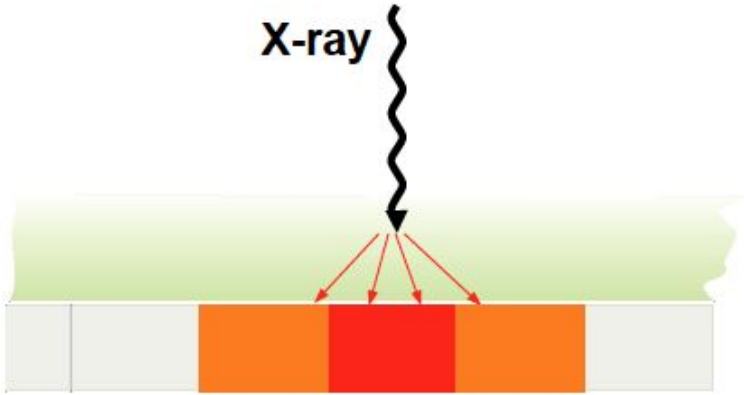
Point Spread Function : 1Pixel

PAD sensor



Difference to CCD

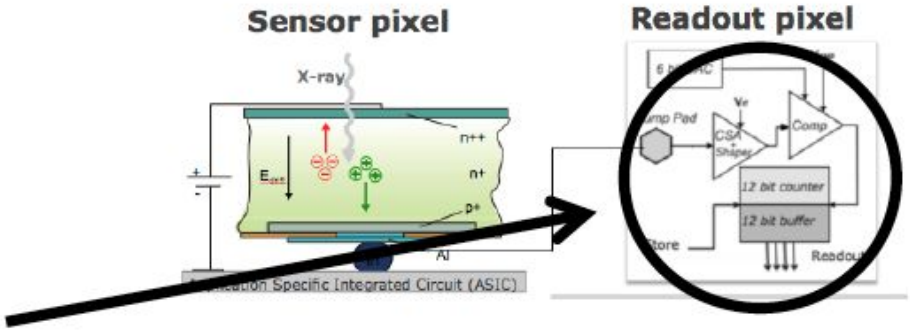
CCD



Energy adjustments

Point 1

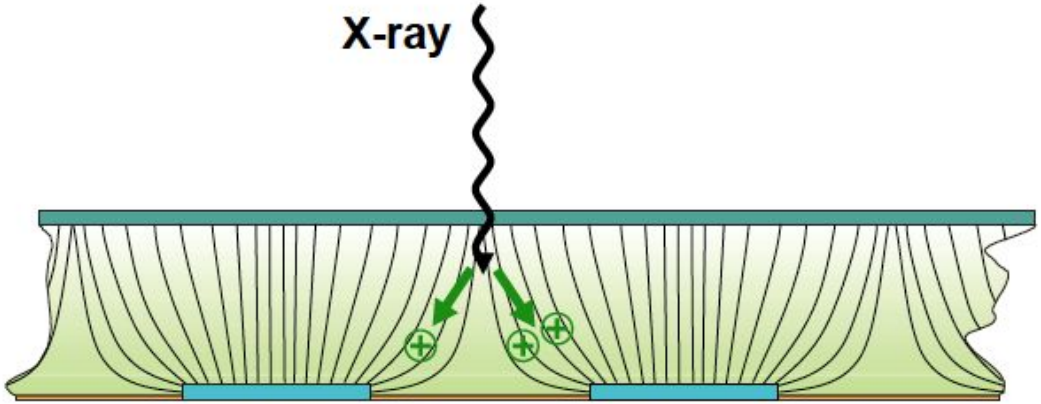
Necessary adjustments to correct for differences



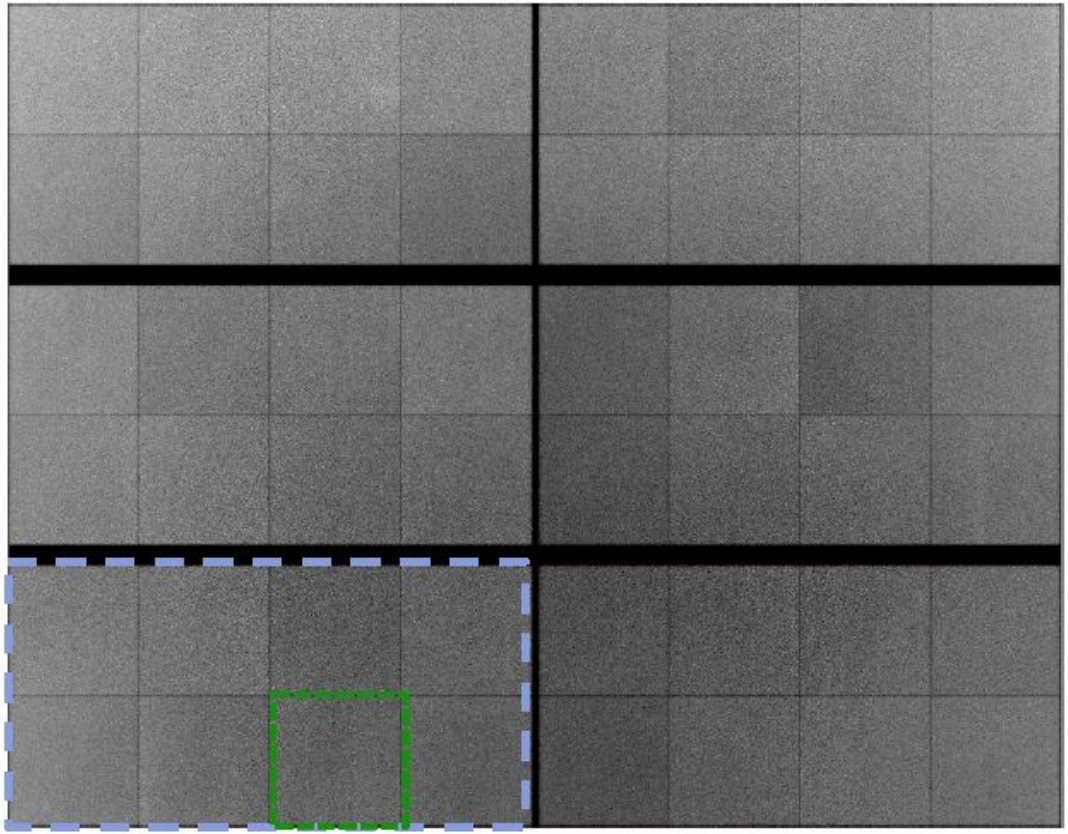
Point 2

Avoiding double counts

=> Threshold ≥ 50%



Effect of Energy Calibration



Un-calibrated detector

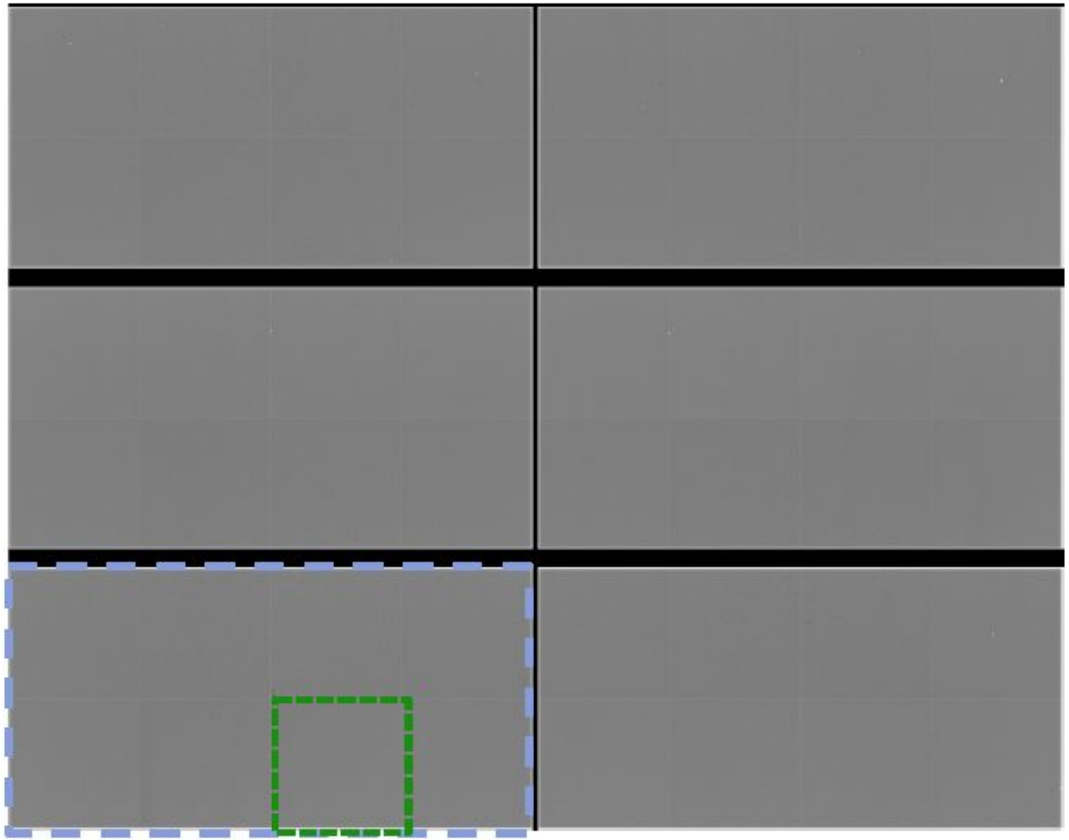
All comparator voltages and trim bits are the same

Differences from chip to chip

Differences within a chip

11.9keV X-rays

Effect of Energy Calibration



Calibrated detector

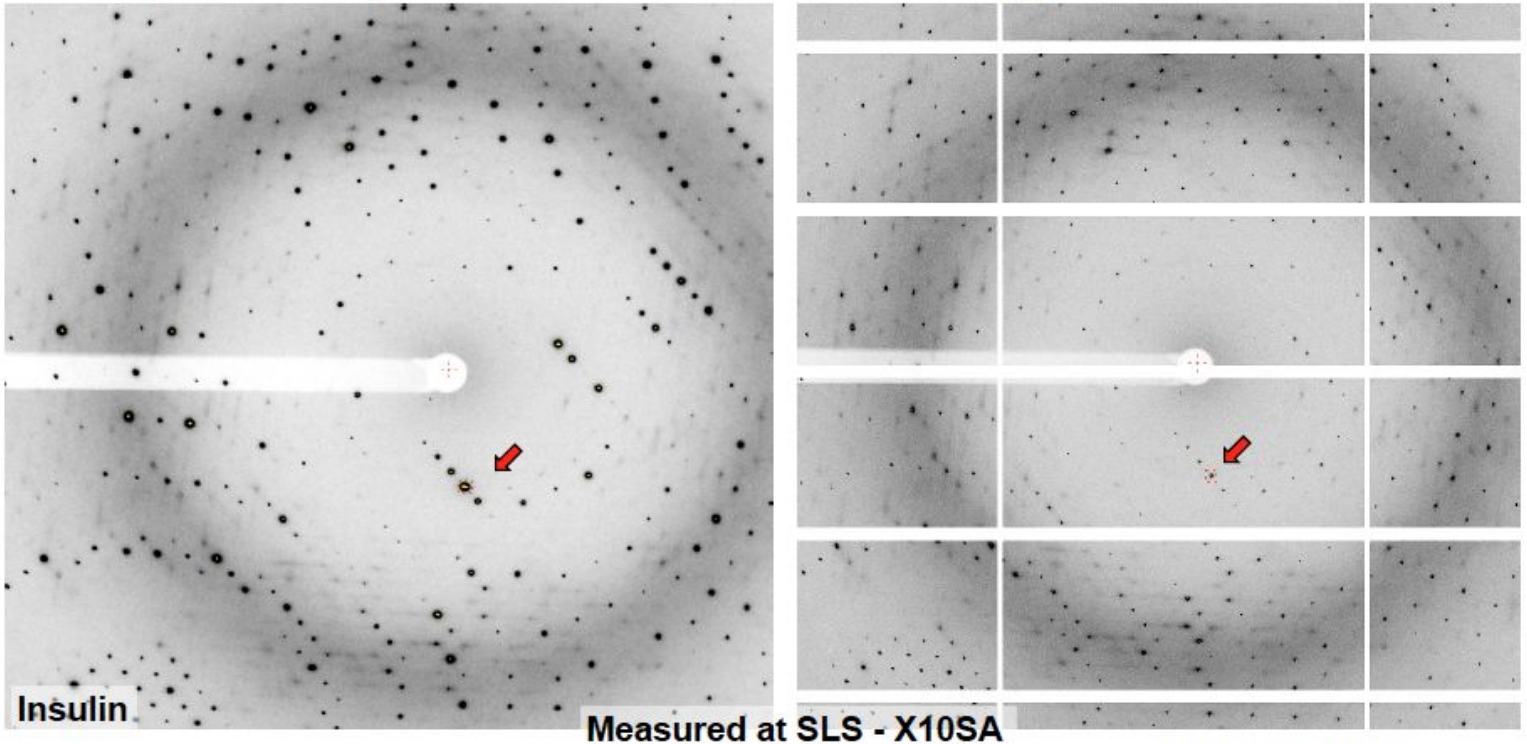
All comparator voltages and trim bits are adjusted that each pixel has a perfect energy calibration

11.9keV X-rays

Crystallography: Dynamic Range and Point Spread Function

CCD
MX-225 marresearch
16 bit (65 535 ADU)

PILATUS
Pilatus 6M
20 bit (1 048 575 counts)

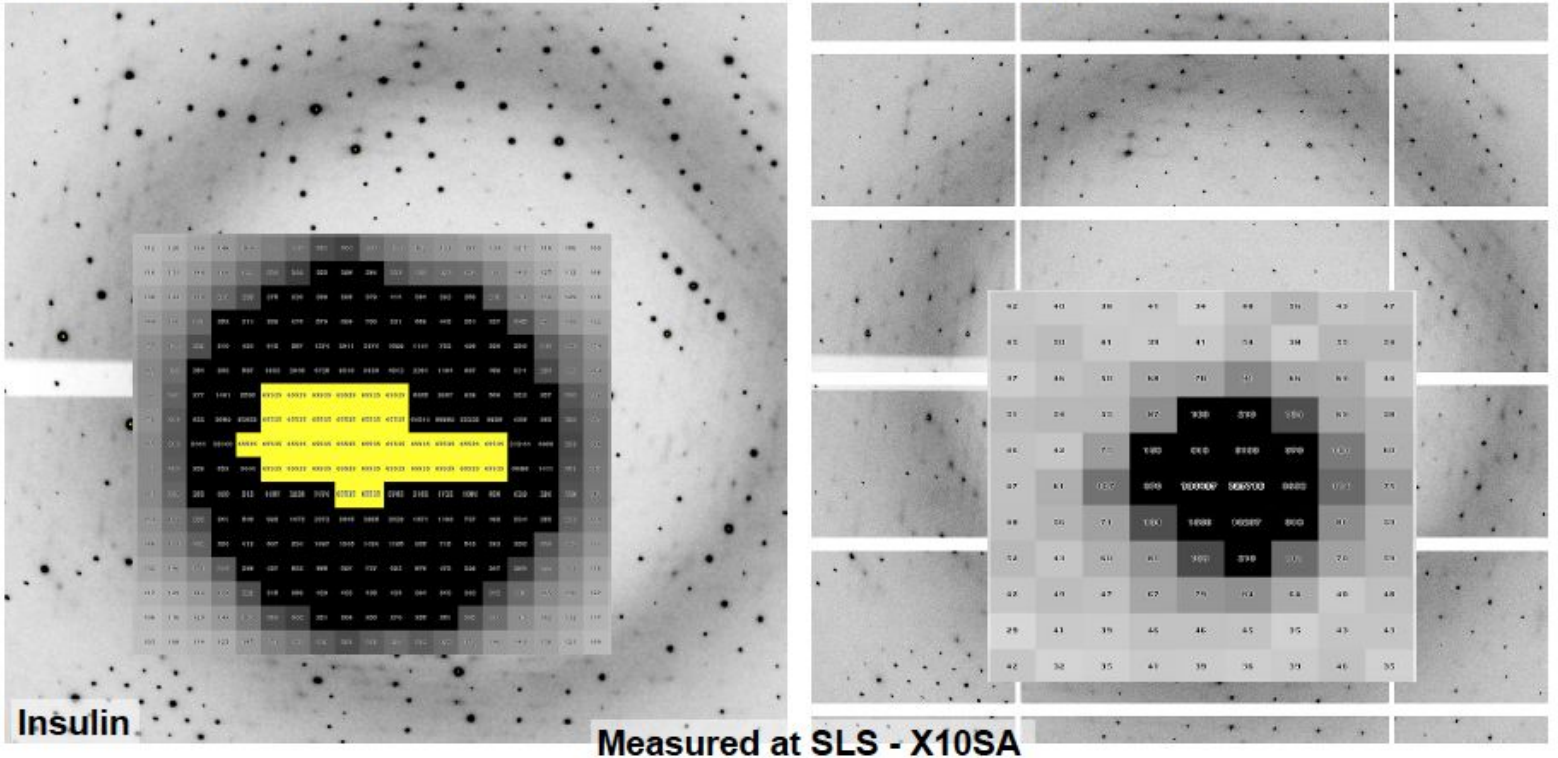


Measured at SLS - X10SA

Crystallography: Dynamic Range and Point Spread Function

CCD
MX-225 marresearch
16 bit (65 535 ADU)

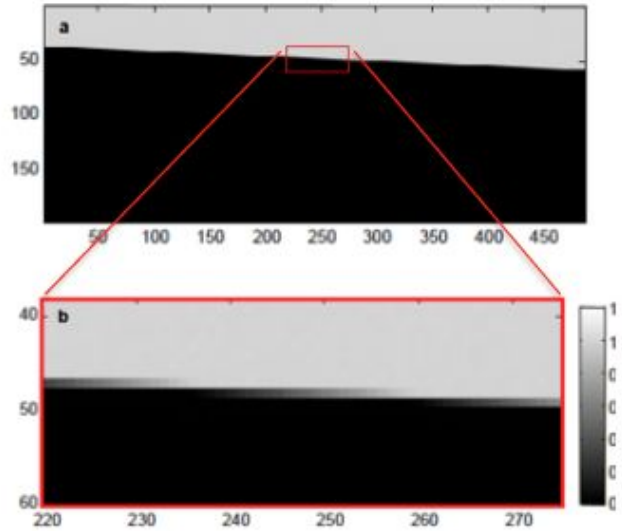
PILATUS
Pilatus 6M
20 bit (1 048 575 counts)



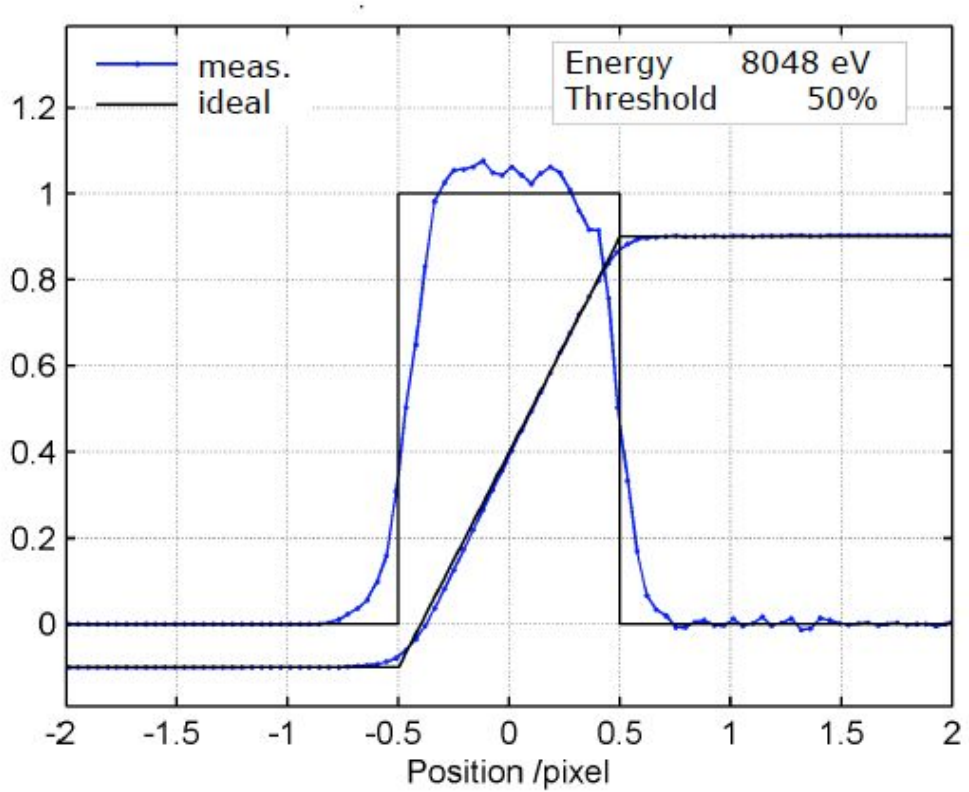
EIGER – spatial resolution

Edge Spread Function

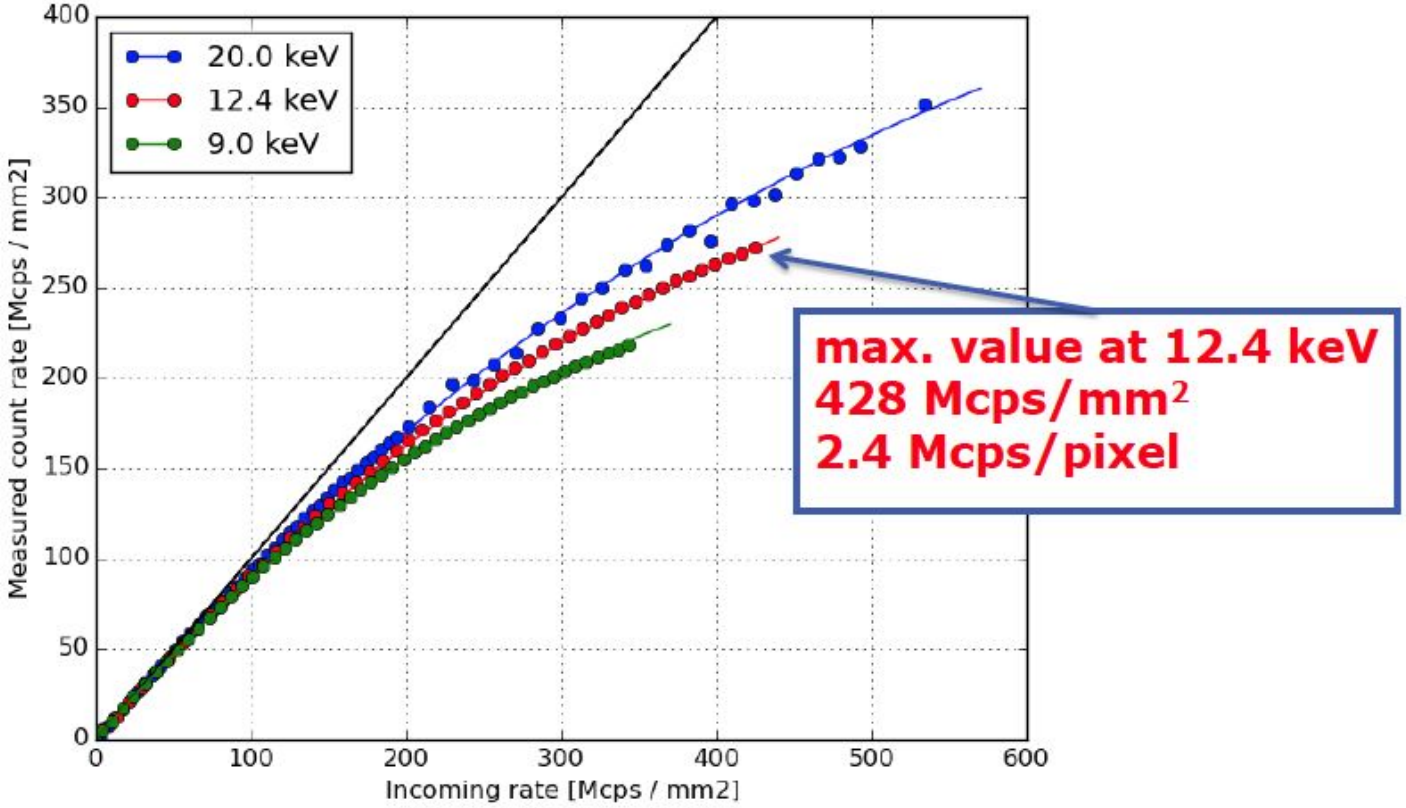
by slanted edge method



Derivative (ESF) => LSF



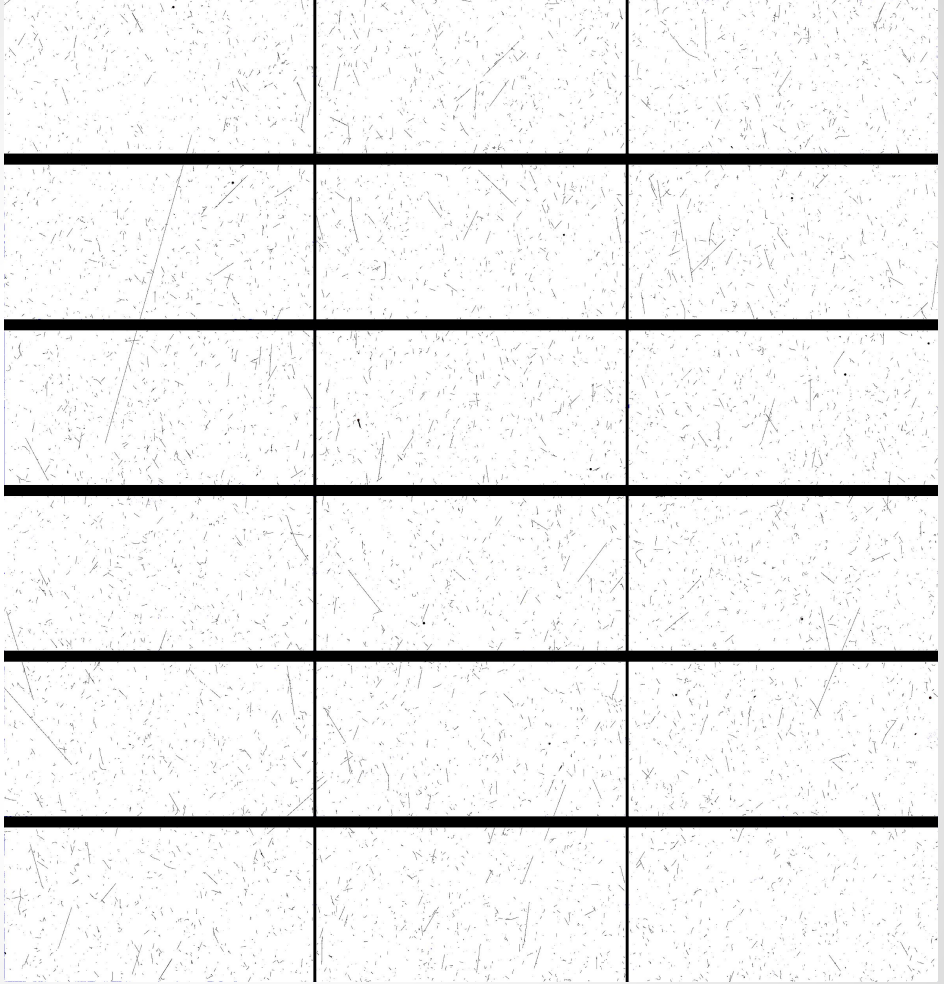
Count rate performance excellent at all energies



	max. frame rate [Hz]	# of images at max. frame rate	continuous frame rate [Hz]
EIGER X 1M	3000	> 90’000	1500
EIGER X 4M	750	> 22’000	375
EIGER X 9M	238	> 7’100	119
EIGER X 16M	133	> 4’000	66

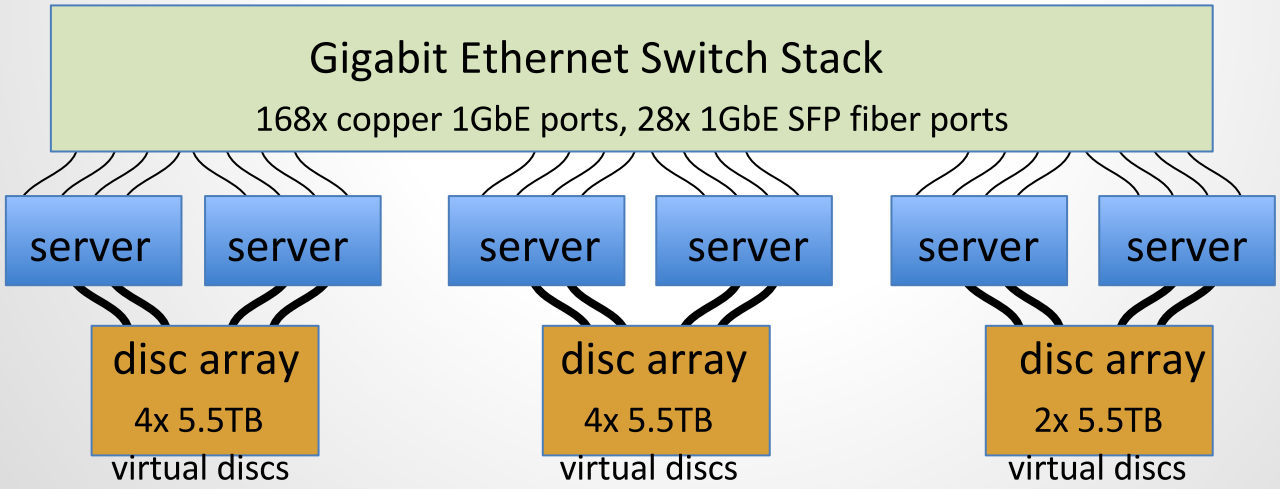
Simple test of our detector's functionality:

- 15 minute exposure
- Cosmic ray traces
- White pixels have zero counts
- Black pixels have one count
- Higgs boson not detected



Data Storage & Network Upgrade

- Lustre distributed file system
  - multiple servers distribute files across virtual discs
  - very stable
- Current disc arrays
  - 55TB in RAID 6 with hot spare (can lose 3 drives per group of 14)
  - SATA drives (inexpensive, large and relatively slow)
- Current servers
  - three failover pairs
  - redundant 3Gb/s connections to disc arrays
  - 4x 1GbE bonded copper connection to network
- Current network switch
  - gigabit ethernet over copper
  - network lines are all CAT6A and capable of 10G over copper



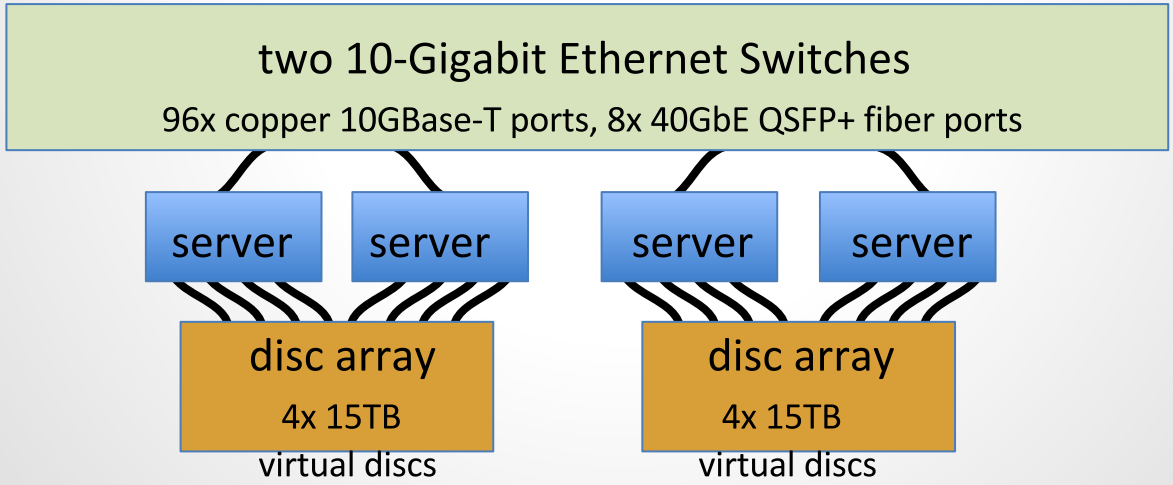
Upgraded Components

keep current system

upgrade all components

keep system adaptable

- New disc arrays
  - near line SAS drives
  - 120TB in RAID 6 (can lose 2 drives per group of 10)
  - only using ¼ capacity
  - can add in solid state, SAS or SATA drives
- New servers
  - multiple 12Gb/s connections to disc array
  - 40GbE fiber connection to switch
- New switches
  - two switches for redundancy
  - most connections copper 10GBase-T (6-7Gb/s)
  - four 40GbE fiber ports on each



Data Collection Software Updates

On our side:

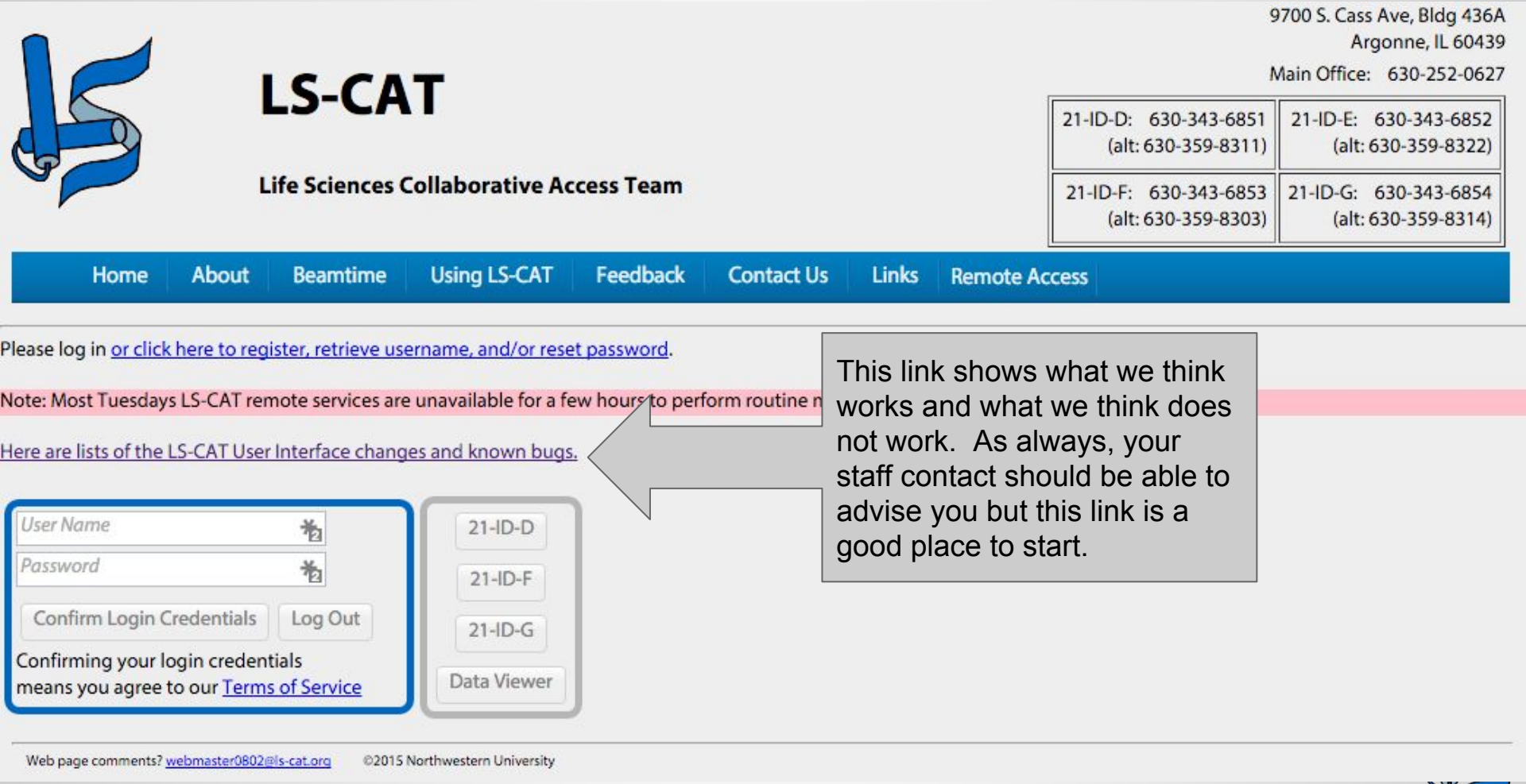
- Incremental improvements to the user experience are being made:
  - Inclusion of diffraction images in the interface to reduce the reliance on our old (but excellent) Adobe Flash interface.
  - Further rollout of the “Loop Explorer” mode (more on following slides).
  - Many minor improvements leading to ease of use and greater reliability.
  - Still many little but annoying bugs.

Data Collection Software Updates

On your side:

- A big problem appears to be the continued use of old browsers.
- We see some problems from old, slow computers without much memory.
- Invest a little time to keep your operating systems up to date.
- We test our software on Mac, Linux, and Windows using, as appropriate, Chrome, Chromium, Edge, Firefox, Opera, and Safari. When you have a problem we'll expect you to first upgrade your current browser or perhaps switch to another.

All recent versions of the above browsers work (although Firefox is "clunky" on all platforms). However, a year ago only Chrome on a Mac worked. Most the improvements have come from the browser developers themselves, not from anything LS-CAT has done.


LS-CAT
Life Sciences Collaborative Access Team
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Argonne, IL 60439
Main Office: 630-252-0627
21-ID-D: 630-343-6851 (alt: 630-359-8311)
21-ID-E: 630-343-6852 (alt: 630-359-8322)
21-ID-F: 630-343-6853 (alt: 630-359-8303)
21-ID-G: 630-343-6854 (alt: 630-359-8314)
Home
About
Beamtime
Using LS-CAT
Feedback
Contact Us
Links
Remote Access
Please log in or click here to register, retrieve username, and/or reset password.
Note: Most Tuesdays LS-CAT remote services are unavailable for a few hours to perform routine n
Here are lists of the LS-CAT User Interface changes and known bugs.
User Name
Password
Confirm Login Credentials
Log Out
Confirming your login credentials means you agree to our Terms of Service
21-ID-D
21-ID-F
21-ID-G
Data Viewer
This link shows what we think works and what we think does not work. As always, your staff contact should be able to advise you but this link is a good place to start.
Web page comments? webmaster0802@ls-cat.org
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Data Collection Software Updates

Chrome 46.0.2490.71 m	Windows 8.1	
Chrome 46.0.2490.71 (64-bit)	Mac OS X 10.11	
Opera 32.0.1948.69	Windows 8.1	
Opera 32.0.1948.69	Mac OS X 10.11	
Firefox 41.0.2	Mac OS X 10.11	Centering Crosshairs “Clunky”
Firefox 41.0.2	Windows 8.1	Centering Crosshairs “Clunky”
Firefox 41.0.2	Ubuntu 14.04 LTS 64-bit	Centering Crosshairs “Clunky”
Safari 9.0 (11601.1.56)	Mac OS X 10.11	
Chromium 45.0.2454.101	Ubuntu 14.04 LTS 64-bit	
Chrome 46.0.2490.71-1	Ubuntu 14.04 LTS 64-bit	
Edge 20.10240.16384.0	Windows 10	Camera menu buttons are way too big but still work just fine

LS-CAT
Life Sciences Collaborative Access Team

e136369

Schedule | 21-ID-D

136369 Brister
Log Out ESAF

You Control This Station
Cede

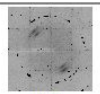
91.0 mA
11668 Cts

Open FE
Open Stn
Close Stn

Manually Mounted Sample

Energy (KeV)
10.000
10.000

Attenuation
Actual
74.10%
Requested
75 %


o.009

Chat

Sample | Center | Collect | Expert

Centering Mode
Loop Explorer

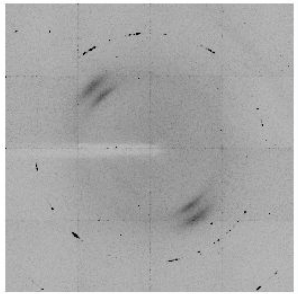
MD2 Mode
Data Collection

Loop Explorer

Freeze | Thaw

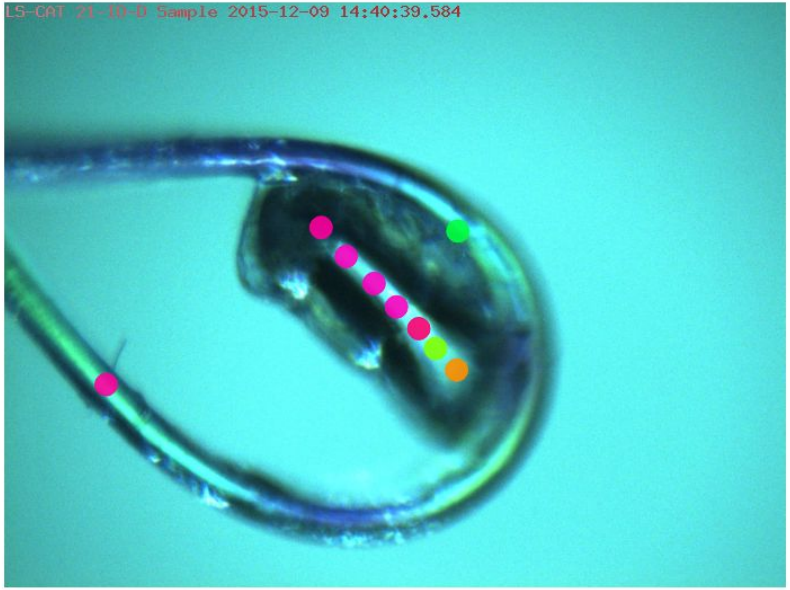
Omega 142.89
Home | -180 | -90 | -10 | +10 | +90 | +180

Directory: d/kb/151209/o
Prefix: o
Distance (mm): 250
Time (Secs): 1
Omega (°): 143
Delta (°): 1


o.009
Follow Data Collection

File Name	Start (°)	Status	Frame	Spots
o.001	142.000	Done	1	38
o.002	142.000	Done	2	36
o.003	142.000	Done	3	36
o.004	142.000	Done	4	36
o.005	142.000	Done	5	72
o.006	142.000	Done	6	57
o.007	142.000	Done	7	41
o.008	142.000	Done	8	38
o.009	142.000	Done	9	84

D Loop Explorer



LS-CAT
Life Sciences Collaborative Access Team

e141036

Schedule | 21-ID-D

141036 Geiger
Log Out ESAF

You Control This Station
Cede
ESAF using this station

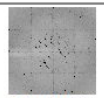
88.5 mA
26007 Cts

Open FE
Open Stn
Close Stn

Dewar: 1
Puck: 2
Sample: 2

Energy (KeV)
10.000
10.000

Attenuation
Actual
41.70%
Requested
40 %



Chat

Sample | Center | Collect | Expert

Centering Mode
Loop Explorer

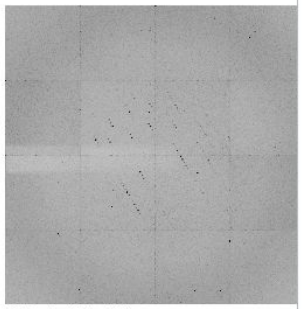
MD2 Mode
Centering Zoom 1

Loop Explorer

Freeze | Thaw

Omega -29.999
Home | -180 | -90 | -10 | +10 | +90 | +180

Directory: d/LE
Prefix: biow_1_2_2
Distance (mm): 450 Time (Secs): 1
Omega (°): -89.5 Delta (°): 1

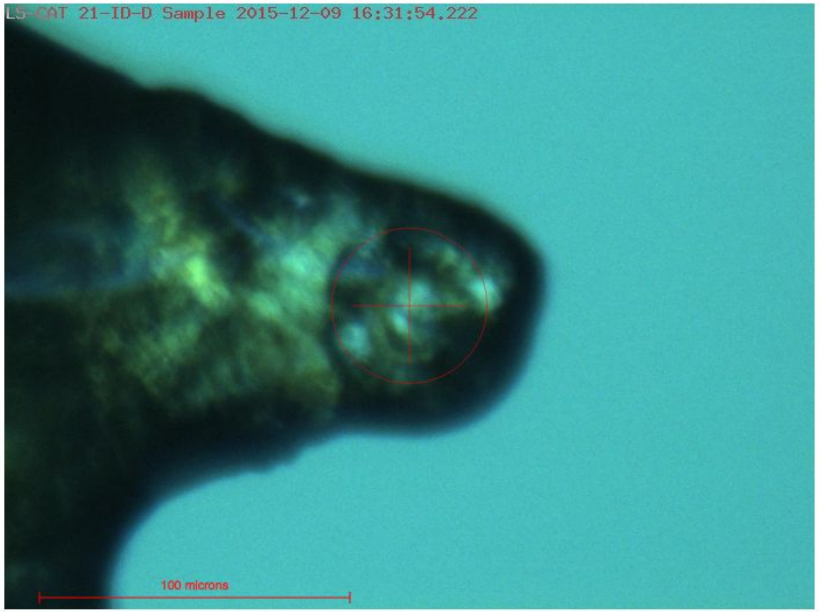


264-p1c.001_006
Follow Data Collection

File Name	Start (°)	Status	Frame
No items to display			

D Sample
Zoom: 1 2 3 4 5 6 7 8 9 10

Replicate Video



LS-CAT
Life Sciences Collaborative Access Team

e141036

Schedule | 21-ID-D

141036 Geiger
Log Out ESAF

You Control This Station
Cede

88.1 mA
34406 Cts

Open FE
Open Stn
Close Stn

Dewar: 1
Puck: 2
Sample: 2

Energy (KeV)
10.000
10.000

Attenuation
Actual
23.70%
Requested
20 %



Chat

Sample | Center | Collect | Expert

Centering Mode
Loop Explorer

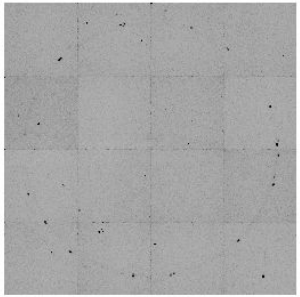
MD2 Mode
Data Collection

Loop Explorer

Freeze | Thaw

Omega -29.604
Home | -180 | -90 | -10 | +10 | +90 | +180

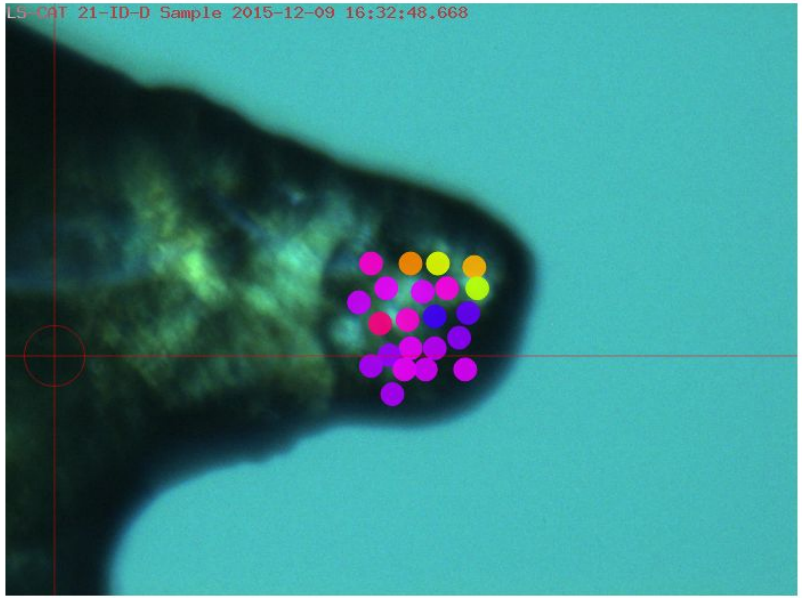
Directory: d/LE
Prefix: 264-p2
Distance (mm): 450
Time (Secs): 1
Omega (°): -30
Delta (°): 1



264-p2.010
Follow Data Collection

File Name	Start (°)	Status	Frame	Spots
264-p2.013	-30.500	Done	13	115
264-p2.014	-30.500	Done	14	129
264-p2.015	-30.500	Done	15	143
264-p2.016	-30.500	Done	16	157
264-p2.017	-30.500	Done	17	132
264-p2.018	-30.500	Done	18	138
264-p2.019	-30.500	Done	19	138
264-p2.020	-30.500	Done	20	159
264-p2.021	-30.500	Done	21	151
264-p2.022	-30.500	Done	22	154

D Loop Explorer



APS-U

- APS-U Call for Beamline Proposals
  - White Paper Due January 25
  - Papers will be evaluated and a subset invited to create a full proposal
  - Unlike CAT proposals, these beamlines would be built and staffed by the APS as part of the upgrade
- The project has not yet been baselined
  - We do not know what the APS-U expects to do, exactly
  - We do not know when the APS-U expects to shut down the APS, exactly
  - My guess of an 18 month down period on or about 2022 appears as good as any. Party line is still 12 months starting in 2021(ish) not 18 months starting 2022(ish)
  - 2016 should see significant progress towards nailing these things down.